

Command Template

L2 (2960 Switch)

```
en
config t
```

```
//untuk setup vlan
vlan [id]
name BxFx
ex
```

```
//untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan [id]
ex
```

```
//untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
//membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan [id],[id] // tanda kuning opsional
ex
```

```
//Opsional (menambah id vlan trunk):
sw trunk allowed vlan add [id]
```

L3 (3650 Multilayer Switch)

```
en
config t
```

```
//untuk mengaktifkan fungsi routing
ip routing
```

```
//untuk setup vlan
vlan [id]
name BxFx
ex
```

```
//untuk mengaitkan vlan dengan default gateway setiap vlan
int vlan [id]
ip addr [default_gateway] [subnet_mask]
no sh
ex
```

```
//untuk setup port GigabitEthernet sebagai trunk downlink ke switch L2 (2960)
int range g1/0/x-y
sw mode trunk
```

```
//membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan [id],[id] // tanda kuning opsional
ex
```

```
//Opsional (menambah id vlan trunk):
sw trunk allowed vlan add [id]
```

//untuk mengatur koneksi L3-Router

int g1/1/1

no switchport

ip addr [L3_Switch_to_Router_via_g1/1/1] [subnet_mask] //lihat di excel sheet L3SW-Router

no sh

ex

//untuk mengatur ospf

router ospf 1

//untuk network L3SW-Router

network [ip] [wildcard] [OSPF_area] //lihat di excel sheet L3SW-Router

//untuk network setiap vlan dalam gedung

network [ip] [wildcard] [OSPF_area] //lihat di excel sheet Mapping

ex

L3 ACL

en
config t

! buat extended access list
ip access-list extended [nama_acl]

! mengizinkan koneksi TCP established (koneksi balasan dari yang di inisiasi pc dest)
permit tcp [[src NA] [src submask] / any] [dest NA] [dest submask] established

! mengizinkan reply ping (ping balasan dari pc source masuk ke dalam dest)
permit [[src NA] [src submask] / any] [dest NA] [dest submask] echo-reply

! mengizinkan koneksi UDP port DNS
permit udp any eq domain [dest NA] [dest submask]

! Jalur khusus untuk balasan DHCP Server
permit udp any eq bootps any eq bootpc

! mengizinkan Finance akses HR
permit ip [[src NA] [src submask] / any] [[dest NA] [dest submask] / any]

! blokir yang lainnya (meng eksplisitkan implicit deny)
deny ip any any
exit

! implementasi (dari multilayer ke subnet) vlan
int vlan [id]
ip access-group [nama_acl] [in/out]
exit
exit
exit

Cara hapus ACL:

Metode 1. Hapus keseluruhan

! Menghapus ACL bernama PROTECT_FINANCE secara keseluruhan
en
config t
no ip access-list extended PROTECT_FINANCE

! Menghapus binding ACL bernama PROTECT_FINANCE
en
config t
int vlan 311
no ip access-group PROTECT_FINANCE out
exit

Metode 2. hapus aturan tertentu

en
config t

Langkah 1: Lihat ACL yang terdaftar

do show access-lists

Outputnya akan terlihat seperti ini:

```
Extended IP access list PROTECT_FINANCE
 10 permit tcp any 172.16.30.0 0.0.0.255 established
 20 permit icmp any 172.16.30.0 0.0.0.255 echo-reply
 30 deny ip 172.16.0.0 0.0.255.255 172.16.30.0 0.0.0.255
```

Langkah 2: Masuk ke dalam mode edit ACL tersebut

ip access-list extended PROTECT_FINANCE

Langkah 3: Hapus baris yang tidak diinginkan menggunakan nomor urut Misalnya
Misalnya ingin menghapus aturan pemblokiran IP (nomor 30):

no 30
exit

Setelah dihapus, silahkan ketikkan aturan permit atau deny yang baru. Aturan baru tersebut secara otomatis akan ditaruh di urutan paling bawah.

Router (ISR 4331)

en
config t

```
//untuk mengatur koneksi L3-Router
int g0/0/1
media-type sfp
ip addr [Router_to_L3_Switch_via_g0/0/0] [subnet_mask] //lihat di excel sheet L3SW-Router
no sh
ex
```

```
//untuk mengatur koneksi antar-router (1/2)
int g0/0/x
ip addr [ip] [subnet_mask] //lihat di excel sheet Inter-Router untuk pasangan routernya
no sh
! apply ospf key and pass in physical interface
ip ospf message-digest-key [nomor_key] md5 [pass]
ex
```

```
//untuk mengatur koneksi antar-router (2/2)
int g0/0/x
ip addr [ip] [subnet_mask] //lihat di excel sheet Inter-Router untuk pasangan routernya
no sh
! apply ospf key and pass in physical interface
ip ospf message-digest-key [nomor_key] md5 [pass]
ex
```

```
//untuk mengatur ospf
router ospf 1
area 0 authentication message-digest
```

```
//untuk network antar-router (1/2)
network [ip] [wildcard] [OSPF_area] //lihat di excel sheet Inter-Router
```

```
//untuk network antar-router (2/2)
network [ip] [wildcard] [OSPF_area] //lihat di excel sheet Inter-Router
```

```
//untuk network L3SW-Router
network [ip] [wildcard] [OSPF_area] //lihat di excel sheet L3SW-Router
ex
```

cara cek ospf md5 auth aktif apa kagak
show ip ospf interface

Cara CEK:
CEK BAWAH!!! ARGGGGH

```
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.10.6/30, Area 0
Process ID 1, Router ID 192.168.10.10, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.10.10, Interface address 192.168.10.6
Backup Designated Router (ID) 192.168.10.5, Interface address 192.168.10.5
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:09
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
Adjacent with neighbor 192.168.10.5 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
Message digest authentication enabled
Youngest key id is 1
GigabitEthernet0/0/2 is up, line protocol is up
Internet address is 192.168.10.10/30, Area 0
Process ID 1, Router ID 192.168.10.10, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.10.10, Interface address 192.168.10.10
Backup Designated Router (ID) 192.168.10.9, Interface address 192.168.10.9
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:09
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
Adjacent with neighbor 192.168.10.9 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
Message digest authentication enabled
Youngest key id is 1
```

Command

PC

Desktop > IP Configuration

Default gateway di suatu network pakai usable ip yang pertama

PC pertama di suatu network pakai usable ip yang kedua

PC kedua di suatu network pakai usable ip yang ketiga

dan seterusnya

L2 Setting

Building A - Floor 1

```
! untuk setup vlan
vlan 11
name BAF1
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 11
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 11
ex
ex
ex
```

Building A - Floor 2

```
! untuk setup vlan
vlan 12
name BAF2
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 12
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 12
ex
ex
ex
```

Building A - Floor 3

```
! untuk setup vlan
vlan 13
name BAF3
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 13
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 13
ex
ex
ex
```

Building A - Floor 4

```
! untuk setup vlan
vlan 14
name BAF4
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 14
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 14
ex
ex
ex
```

Building B - Floor 1

```
! untuk setup vlan
vlan 21
name BBF1
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 21
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 21
ex
ex
ex
```

Building B - Floor 2

```
! untuk setup vlan
vlan 22
name BBF2
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 22
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 22
ex
ex
ex
```

Building B - Floor 3

```
! untuk setup vlan
vlan 23
name BBF3
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 23
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 23
ex
ex
ex
```

Building B - Floor 4

```
! untuk setup vlan
vlan 24
name BBF4
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 24
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 24
ex
ex
ex
```

Building C - Floor 1 FIN

```
! untuk setup vlan  
vlan 311  
name BCF1-FIN  
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id  
int range fa0/1-24  
sw mode access  
sw access vlan 311  
ex
```

```
! untuk setup uplink ke switch L3 (3650)  
int g0/1  
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:  
sw trunk allowed vlan 311  
ex  
ex  
ex
```

Building C - Floor 1 IT

```
! untuk setup vlan  
vlan 312  
name BCF1-IT  
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id  
int range fa0/1-24  
sw mode access  
sw access vlan 312  
ex
```

```
! untuk setup uplink ke switch L3 (3650)  
int g0/1  
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:  
sw trunk allowed vlan 312  
ex  
ex  
ex
```

Building C - Floor 1 TEC

```
! untuk setup vlan  
vlan 313  
name BCF1-TEC
```

ex

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 313
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 313
ex
ex
ex
```

Building C - Floor 1 HR

```
! untuk setup vlan
vlan 314
name BCF1-HR
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
sw mode access
sw access vlan 314
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 314
ex
ex
ex
```

Building C - Floor 2

```
! untuk setup vlan
vlan 32
name BCF2
ex
```

```
! untuk atur port FastEthernet sebagai access dan mengaitkannya dengan vlan id
int range fa0/1-24
```

```
sw mode access
sw access vlan 32
ex
```

```
! untuk setup uplink ke switch L3 (3650)
int g0/1
sw mode trunk
```

```
! membatasi hanya vlan id ini yang bisa lewat trunk:
sw trunk allowed vlan 32
ex
ex
ex
```

Port Security

GEDUNG A

Switch BAF1 (g1/0/1 di Multilayer A, VLAN 11)

```
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 11
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
```

ex

ex

Switch BAF2 (g1/0/4 di Multilayer A, VLAN 12)

```
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 12
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
```

ex

ex

Switch BAF3 (g1/0/8 di Multilayer A, VLAN 13)

```
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 13
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
```

ex

ex

Switch BAF4 (g1/0/10 di Multilayer A, VLAN 14)

```
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 14
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
```

ex

ex

GEDUNG B

Switch BBF1 (g1/0/1 di Multilayer B, VLAN 21)

en

config t

int range fa0/1-24

switchport mode access

switchport access vlan 21

switchport port-security

switchport port-security maximum 1

switchport port-security mac-address sticky

switchport port-security violation shutdown

ex

ex

Switch BBF2 (g1/0/6 di Multilayer B, VLAN 22)

en

config t

int range fa0/1-24

switchport mode access

switchport access vlan 22

switchport port-security

switchport port-security maximum 1

switchport port-security mac-address sticky

switchport port-security violation shutdown

ex

ex

Switch BBF3 (g1/0/10 di Multilayer B, VLAN 23) — *dari mapping: g1/0/10*

en

config t

int range fa0/1-24

switchport mode access

switchport access vlan 23

switchport port-security

switchport port-security maximum 1

switchport port-security mac-address sticky

switchport port-security violation shutdown

ex

ex

Switch BBF4 (g1/0/11 di Multilayer B, VLAN 24)

en

config t

int range fa0/1-24

switchport mode access

switchport access vlan 24

switchport port-security

switchport port-security maximum 1

switchport port-security mac-address sticky

```
switchport port-security violation shutdown
ex
ex
```

GEDUNG C

Switch BCF1-FIN (g1/0/1 di Multilayer C, VLAN 311)

```
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 311
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
ex
ex
```

Switch BCF1-IT (g1/0/6 di Multilayer C, VLAN 312)

```
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 312
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
ex
ex
```

Switch BCF1-TEC (g1/0/9 di Multilayer C, VLAN 313)

```
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 313
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
ex
ex
```

Switch BCF1-HR (g1/0/11 di Multilayer C, VLAN 314)

```
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 314
switchport port-security
```

```
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
ex
ex
Switch BCF2 (g1/0/12 di Multilayer C, VLAN 32)
en
config t
int range fa0/1-24
switchport mode access
switchport access vlan 32
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
ex
ex
```

DHCP Server L2 config

NOTE: PASTIKAN sudah setting di L3 terlebih dahulu (L3 (Multilayer) Setting > DHCP Server L3 config)!

1. Mendaftarkan fisik VLAN tersebut di bagian Access Switch. Number dan Nama VLAN mengikuti VLAN Database di MultiLayer Switch.

```
Switch(config)# vlan [number]
Switch(config-vlan)# name [name]
```

2. Masukkan Port PC ke dalam VLAN

```
Switch(config)# interface [interface_ethernet]
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan [number]
```

Catatan:

Mulai dari langkah nomor 2 hingga 5, itu berlaku untuk satu PC dan satu Access Switch saja. Sehingga untuk PC dan Switch yang lain, langkah 2 hingga 5 harus diulang, dengan *excluded-address*, *pool*, *network*, *default-router*, *vlan number*, *vlan name*, dan *interface Switch* menyesuaikan setiap PC.

L3 (Multilayer) Setting

Multilayer A:

```
! enable routing
ip routing
```

```
! vlan creation
vlan 11
name BAF1
ex
```

```
vlan 12
name BAF2
ex
```

```
vlan 13
name BAF3
ex
```

```
vlan 14
name BAF4
ex
```

```
! vlan linking
int vlan 11
ip addr 172.16.10.1 255.255.255.128
no sh
ex
```

```
int vlan 12
ip addr 172.16.10.129 255.255.255.128
no sh
ex
```

```
int vlan 13
ip addr 172.16.11.1 255.255.255.192
no sh
ex
```

```
int vlan 14
ip addr 172.16.11.65 255.255.255.240
no sh
ex
```

```
! vlan trunk downlink
int range g1/0/1-3
sw mode trunk
sw trunk allowed vlan 11
ex
```

```
int range g1/0/4-7
sw mode trunk
sw trunk allowed vlan 12
ex
```

```
int range g1/0/8-9
sw mode trunk
sw trunk allowed vlan 13
ex
```

```
int g1/0/10
sw mode trunk
sw trunk allowed vlan 14
ex
```

```
! convert 1 switch port to routing port
int g1/1/1
no switchport
ip addr 172.16.1.2 255.255.255.252
no sh
ex
```

```
! ospf
router ospf 1
```

```
! building A multilayer-router network (ospf area 10)
network 172.16.1.0 0.0.0.3 area 10
```

```
! building A floor networks (ospf area 10)
network 172.16.10.0 0.0.0.127 area 10
network 172.16.10.128 0.0.0.127 area 10
network 172.16.11.0 0.0.0.63 area 10
network 172.16.11.64 0.0.0.15 area 10
ex
ex
ex
```

Multilayer B:

```
! enable routing  
ip routing
```

```
! vlan creation  
vlan 21  
name BBF1  
ex
```

```
vlan 22  
name BBF2  
ex
```

```
vlan 23  
name BBF3  
ex
```

```
vlan 24  
name BBF4  
ex
```

```
! vlan linking  
int vlan 21  
ip addr 172.16.20.1 255.255.255.0  
no sh  
ex
```

```
int vlan 22  
ip addr 172.16.21.1 255.255.255.128  
no sh  
ex
```

```
int vlan 23  
ip addr 172.16.21.129 255.255.255.224  
no sh  
ex
```

```
int vlan 24  
ip addr 172.16.21.161 255.255.255.240  
no sh  
ex
```

```
! vlan trunk downlink  
int range g1/0/1-5  
sw mode trunk  
sw trunk allowed vlan 21  
ex
```

```
int range g1/0/6-9
sw mode trunk
sw trunk allowed vlan 22
ex
```

```
int g1/0/10
sw mode trunk
sw trunk allowed vlan 23
ex
```

```
int g1/0/11
sw mode trunk
sw trunk allowed vlan 24
ex
```

```
! convert 1 switch port to routing port
int g1/1/1
no switchport
ip addr 172.16.2.2 255.255.255.252
no sh
ex
```

```
! ospf
router ospf 1
```

```
! building B multilayer-router network (ospf area 20)
network 172.16.2.0 0.0.0.3 area 20
```

```
! building B floor networks (ospf area 20)
network 172.16.20.0 0.0.0.255 area 20
network 172.16.21.0 0.0.0.127 area 20
network 172.16.21.128 0.0.0.31 area 20
network 172.16.21.160 0.0.0.15 area 20
ex
ex
ex
```

Multilayer C:

```
! enable routing  
ip routing
```

```
! vlan creation  
vlan 311  
name BCF1-FIN  
ex
```

```
vlan 312  
name BCF1-IT  
ex
```

```
vlan 313  
name BCF1-TEC  
ex
```

```
vlan 314  
name BCF1-HR  
ex
```

```
vlan 32  
name BCF2  
ex
```

```
! vlan linking  
int vlan 311  
ip addr 172.16.30.1 255.255.255.0  
no sh  
ex
```

```
int vlan 312  
ip addr 172.16.31.1 255.255.255.128  
no sh  
ex
```

```
int vlan 313  
ip addr 172.16.31.129 255.255.255.192  
no sh  
ex
```

```
int vlan 314  
ip addr 172.16.31.193 255.255.255.224  
no sh  
ex
```

```
int vlan 32  
ip addr 172.16.32.1 255.255.255.0
```

```
no sh
ex
```

```
! vlan trunk downlink
int range g1/0/1-5
sw mode trunk
sw trunk allowed vlan 311
ex
```

```
int range g1/0/6-8
sw mode trunk
sw trunk allowed vlan 312
ex
```

```
int range g1/0/9-10
sw mode trunk
sw trunk allowed vlan 313
ex
```

```
int g1/0/11
sw mode trunk
sw trunk allowed vlan 314
ex
```

```
int range g1/0/12-16
sw mode trunk
sw trunk allowed vlan 32
ex
```

```
! convert 1 switch port to routing port
int g1/1/1
no switchport
ip addr 172.16.3.2 255.255.255.252
no sh
ex
```

```
! ospf
router ospf 1
```

```
! building B multilayer-router network (ospf area 30)
network 172.16.3.0 0.0.0.3 area 30
```

```
! building B floor networks (ospf area 30)
network 172.16.30.0 0.0.0.255 area 30
network 172.16.31.0 0.0.0.127 area 30
network 172.16.31.128 0.0.0.63 area 30
network 172.16.31.192 0.0.0.31 area 30
network 172.16.32.0 0.0.0.255 area 30
```

ex
ex
ex

L3 ACL Setting

Reapply ACL Binding (implemented in Multilayer C):

! implementasi ulang outbound (dari multilayer ke subnet) vlan finance (311)

int vlan 311

no ip access-group PROTECT_FINANCE out

ip access-group PROTECT_FINANCE out

exit

! implementasi ulang outbound (dari multilayer ke subnet) vlan hr (314)

int vlan 314

no ip access-group PROTECT_HR out

ip access-group PROTECT_HR out

exit

exit

exit

ACL to protect Finance (implemented in Multilayer C):

! buat extended access list

ip access-list extended PROTECT_FINANCE

! mengizinkan koneksi TCP established (koneksi balasan dari yang di inisiasi pc finance)

permit tcp any 172.16.30.0 0.0.0.255 established

! mengizinkan reply ping (ping balasan dari pc di luar finance masuk ke dalam finance)

permit icmp any 172.16.30.0 0.0.0.255 echo-reply

! mengizinkan koneksi udp port DNS

permit udp any eq domain 172.16.30.0 0.0.0.255

! Jalur khusus untuk balasan DHCP Server

permit udp any eq bootps any eq bootpc

! blokir yang lainnya ("meng eksplisitkan" implicit deny)

deny ip any any

exit

! implementasi outbound (dari multilayer ke subnet) vlan finance (311)

int vlan 311

ip access-group PROTECT_FINANCE out

exit

exit

exit

ACL to protect HR (implemented in Multilayer C):

! buat extended access list

ip access-list extended PROTECT_HR

! mengizinkan koneksi TCP established (koneksi balasan dari yang di inisiasi pc finance)

permit tcp any 172.16.31.192 0.0.0.31 established

! mengizinkan reply ping (ping balasan dari pc di luar finance masuk ke dalam finance)
permit icmp any 172.16.31.192 0.0.0.31 echo-reply

! mengizinkan koneksi UDP port DNS
permit udp any eq domain 172.16.31.192 0.0.0.31

! Jalur khusus untuk balasan DHCP Server
permit udp any eq bootps any eq bootpc

! blokir yang lainnya ("meng eksplisitkan" implicit deny)
deny ip any any
exit

! implementasi outbound (dari multilayer ke subnet) vlan finance (311)
int vlan 314
ip access-group PROTECT_HR out
exit
exit
exit

DHCP Server L3 config

NOTE: Implement 3 bagian ini dulu SEBELUM ke setting DHCP L2 (L2 Setting > DHCP Server L2 config)

1. Allow semua VLAN melewati jalur Trunk di SwitchPort pada MultiLayer Switch

```
Multilayer(config)# interface range [interface_ethernet]
Multilayer(config-if-range)# switchport trunk allowed vlan all
```

2. Kecualikan IP yang sudah ditentukan untuk Gateway (saat mode static), sehingga tidak dibagikan ke PC.

```
Multilayer(config)# ip dhcp excluded-address [ip]
```

3. Buat DHCP Pool untuk sebuah VLAN number, mengikuti daftar di VLAN Database di MultiLayer Switch

```
Multilayer(config)# ip dhcp pool POOL_VLAN[Number]
Multilayer(dhcp-config)# network [ip_network] [subnet]
Multilayer(dhcp-config)# default-router [excluded ip]
```

Catatan:

Mulai dari langkah nomor 2 hingga 5, itu berlaku untuk satu PC dan satu Access Switch saja. Sehingga untuk PC dan Switch yang lain, langkah 2 hingga 5 harus diulang, dengan *excluded-address*, *pool*, *network*, *default-router*, *vlan number*, *vlan name*, dan *interface Switch* menyesuaikan setiap PC.

Router Setting

Router A:

! a to mul a

```
int g0/0/1
ip addr 172.16.1.1 255.255.255.252
no sh
ex
```

! a to c

```
int g0/0/2
ip addr 192.168.10.9 255.255.255.252
no sh
```

```
! apply ospf key and pass in physical interface
ip ospf message-digest-key 1 md5 CN!52026ONEER
ex
```

! a to b

```
int g0/0/0
media-type sfp
ip addr 192.168.10.1 255.255.255.252
no sh
```

```
! apply ospf key and pass in physical interface
ip ospf message-digest-key 1 md5 CN!52026ONEER
ex
```

! setting ospf dan auth nya untuk area 0

```
router ospf 1
area 0 authentication message-digest
```

```
! router A multilayer-router network (ospf area 10)
network 172.16.1.0 0.0.0.3 area 10
```

```
! router A inter-router
network 192.168.10.0 0.0.0.3 area 0
network 192.168.10.8 0.0.0.3 area 0
ex
ex
ex
```

Router B:

! b to mul b

```
int g0/0/1
ip addr 172.16.2.1 255.255.255.252
no sh
ex
```

! b to a

```
int g0/0/0
media-type sfp
ip addr 192.168.10.2 255.255.255.252
no sh
```

! apply ospf key and pass in physical interface

```
ip ospf message-digest-key 1 md5 CN!52026ONEER
ex
```

! b to c

```
int g0/0/2
ip addr 192.168.10.5 255.255.255.252
no sh
```

! apply ospf key and pass in physical interface

```
ip ospf message-digest-key 1 md5 CN!52026ONEER
ex
```

! setting ospf dan auth nya untuk area 0

```
router ospf 1
area 0 authentication message-digest
```

! router B multilayer-router network (ospf area 20)

```
network 172.16.2.0 0.0.0.3 area 20
```

! router B inter-router

```
network 192.168.10.0 0.0.0.3 area 0
network 192.168.10.4 0.0.0.3 area 0
ex
ex
ex
```

Router C:

! c to mul c

```
int g0/0/1
ip addr 172.16.3.1 255.255.255.252
no sh
ex
```

! a to c

```
int g0/0/2
ip addr 192.168.10.10 255.255.255.252
no sh
```

```
! apply ospf key and pass in physical interface
ip ospf message-digest-key 1 md5 CN!52026ONEER
ex
```

! b to c

```
int g0/0/0
media-type sfp
ip addr 192.168.10.6 255.255.255.252
no sh
```

```
! apply ospf key and pass in physical interface
ip ospf message-digest-key 1 md5 CN!52026ONEER
ex
```

! setting ospf dan auth nya untuk area 0

```
router ospf 1
area 0 authentication message-digest
```

```
! router C multilayer-router network (ospf area 30)
network 172.16.3.0 0.0.0.3 area 30
```

```
! router C inter-router
network 192.168.10.4 0.0.0.3 area 0
network 192.168.10.8 0.0.0.3 area 0
ex
ex
ex
```

ABR Command

Router A:

- enable
- configure terminal
- router ospf 1
- area 10 range 172.16.10.0 255.255.254.0
- show ip route ospf (ngecek)

Router B:

- enable
- configure terminal
- router ospf 1
- area 20 range 172.16.20.0 255.255.254.0
- show ip route ospf (ngecek)

Router C:

- enable
- configure terminal
- router ospf 1
- area 30 range 172.16.30.0 255.255.254.0
- area 30 range 172.16.32.0 255.255.255.0
- area 30 range 172.16.32.0 255.255.255.0